



Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Academic Year : 2024-25

Experiment No.3
To perform data cleaning on social media data using python or R.
Date of Performance:01/02/2024
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Aim : To perform data cleaning on social media data using python.

Objective : To analyze data cleaning on social media data using Python is to prepare the data for meaningful analysis, ensure data integrity and quality, and facilitate efficient and ethical use of the data for generating actionable insights.

Theory :

Data cleaning is a crucial step in the data preprocessing pipeline. It involves identifying and correcting errors, inconsistencies, and inaccuracies in the data to improve its quality and reliability. In the context of social media data, which is often unstructured and noisy, data cleaning becomes even more essential.

- **Ensure Data Quality:** The primary objective of data cleaning is to ensure that the data is accurate, consistent, and reliable. Social media data can contain various types of errors such as misspellings, grammatical mistakes, and inconsistencies that need to be addressed.
- **Handle Missing Values:** Social media data often contains missing values due to incomplete user inputs or data collection processes. Data cleaning involves identifying and handling these missing values appropriately, either by imputation or removal.
- **Remove Duplicates:** Social media data may contain duplicate entries, such as duplicate posts or comments. Removing duplicates ensures that each piece of information is unique and prevents redundancy in the dataset.
- **Standardize Formats:** Social media data can have diverse formats for representing dates, times, and other structured information. Data cleaning involves standardizing these formats to facilitate analysis and comparison across different data points.
- **Text Cleaning and Preprocessing:** Since social media data often consists of text data, cleaning and preprocessing text is essential. This may include removing special characters, URLs, hashtags, mentions, and other noise, as well as tokenization, lemmatization, and removing stopwords to prepare the text for analysis.
- **Ensure Consistency and Uniformity:** Data cleaning ensures that the data is consistent and uniform across different attributes and records. This consistency is crucial for accurate analysis and modeling.
- **Enhance Analytical Results:** Clean data leads to more accurate and reliable analytical results. By removing errors and inconsistencies, data cleaning improves the quality of insights derived from social media data analysis.
- **Compliance and Ethical Considerations:** Data cleaning may also involve ensuring compliance with regulations such as GDPR (General Data Protection Regulation) and addressing ethical considerations such as privacy concerns when dealing with sensitive user data in social media datasets.



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Handle Missing Values: Check for missing values and decide how to handle them. Options include dropping rows with missing values, filling them with a default value, or using more sophisticated methods like interpolation.

```
# Drop rows with missing values
data.dropna(inplace=True)
# Fill missing values with a default value
data.fillna(0, inplace=True)
```

Remove Duplicates: Remove any duplicate rows in the dataset.

```
data.drop_duplicates(inplace=True)
```

Text Cleaning: Preprocess text data by removing special characters, URLs, hashtags, mentions, and performing other text cleaning tasks.

```
def clean_text(text):
# Remove URLs
text = re.sub(r'http\S+', '', text)
# Remove special characters and punctuation
text = re.sub(r'^\w\s', '', text)
# Remove numbers
text = re.sub(r'\d+', '', text)
# Convert to lowercase
text = text.lower()
return text
data['clean_text'] = data['text'].apply(clean_text)
```

Tokenization and Lemmatization/Stemming: Tokenize the text and perform lemmatization or stemming to standardize words.

```
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
def tokenize_and_lemmatize(text):
tokens = word_tokenize(text)
lemmatized_tokens = [lemmatizer.lemmatize(token) for token in tokens]
return lemmatized_tokens
data['tokenized_text'] = data['clean_text'].apply(tokenize_and_lemmatize)
```

Program:
`import pandas as pd`
`import numpy as np`

```
file_path = '/content/icc_fb_page.csv'
```



```
# Load the dataset
df = pd.read_csv(file_path)

print("Dataset Overview:")
print(df.info())

print("\nFirst few rows of the dataset:")
print(df.head())

# 1. Handling missing values
# Check for missing values
print("\nMissing values in each column:")
print(df.isnull().sum())

# Drop rows with missing values in the 'text' column
df.dropna(subset=['text'], inplace=True)

# Check if missing values were handled
print("\nMissing values after handling:")
print(df.isnull().sum())

# 2. Remove duplicate rows
# Number of rows initially without cleaning
print("\nNumber of rows before removing duplicates:", df.shape[0])

# Remove duplicate rows
df.drop_duplicates(inplace=True)
print("\nNumber of rows after removing duplicates:", df.shape[0])

# 3. Standardizing text columns (e.g., the 'text' column)
df['text'] = df['text'].str.strip() # Remove extra spaces from the
'text' column
df['text'] = df['text'].str.lower() # Convert the 'text' column to
lowercase

# 4. Handling outliers (if any)
threshold = 1000 # Set your threshold for 'likes'
```



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```
df = df[df['likes'] <= threshold] # Remove rows where 'likes' exceed
the threshold

# Drop columns like 'comments' and any other irrelevant columns
df.drop(columns=['comments', 'shares'], inplace=True) # Adjust
according to your needs

# 7. Check data types after cleaning
print("\nData types after cleaning:")
print(df.dtypes)

# Save the cleaned data to a new CSV file
df.to_csv('/content/icc_fb_page_cleaned.csv', index=False)

print("\nData cleaning complete. Cleaned data saved as
'icc_fb_page_cleaned.csv'.")
```

Output:

```
Dataset Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   comments    2000 non-null   int64
1   likes       2000 non-null   int64
2   shares      2000 non-null   int64
3   text        1996 non-null   object
dtypes: int64(3), object(1)
memory usage: 62.6+ KB
None
```

First few rows of the dataset:

```
   comments  likes  shares
0         161   3594      9  A busy road to ICC Men's #T20WorldCup 2024
for...
1         382  10020   236  Sri Lanka level the series 1-1 after a
stellar...
2          97   2197      4  Dimuth Karunaratne believes Sri Lanka did
well...
3        1605  39370   154  Shakib Al Hasan is back playing Test
cricket a...
4         492  12679   98  Brilliant knocks from Dimuth Karunaratne
and K...
```



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```
Missing values in each column:
```

```
comments    0
likes        0
shares       0
text         4
dtype: int64
```

```
Missing values after handling:
```

```
comments    0
likes        0
shares       0
text         0
dtype: int64
```

```
Number of rows before removing duplicates: 1996
```

```
Number of rows after removing duplicates: 1996
```

```
Data types after cleaning:
```

```
likes        int64
text         object
dtype: object
```

```
Data cleaning complete. Cleaned data saved as
'icc_fb_page_cleaned.csv'.
```

Conclusion: In conclusion, the data cleaning process on social media data is essential for improving data quality, ensuring its integrity, and preparing it for meaningful analysis. By handling missing values, removing duplicates, standardizing text, and addressing outliers, the dataset becomes more reliable for further analysis. Text preprocessing, such as removing special characters and performing tokenization and lemmatization, ensures the data is structured for natural language processing tasks. This experiment demonstrates the importance of cleaning social media data using Python to generate actionable insights and maintain ethical standards, ultimately leading to more accurate and insightful results in social media analytics.